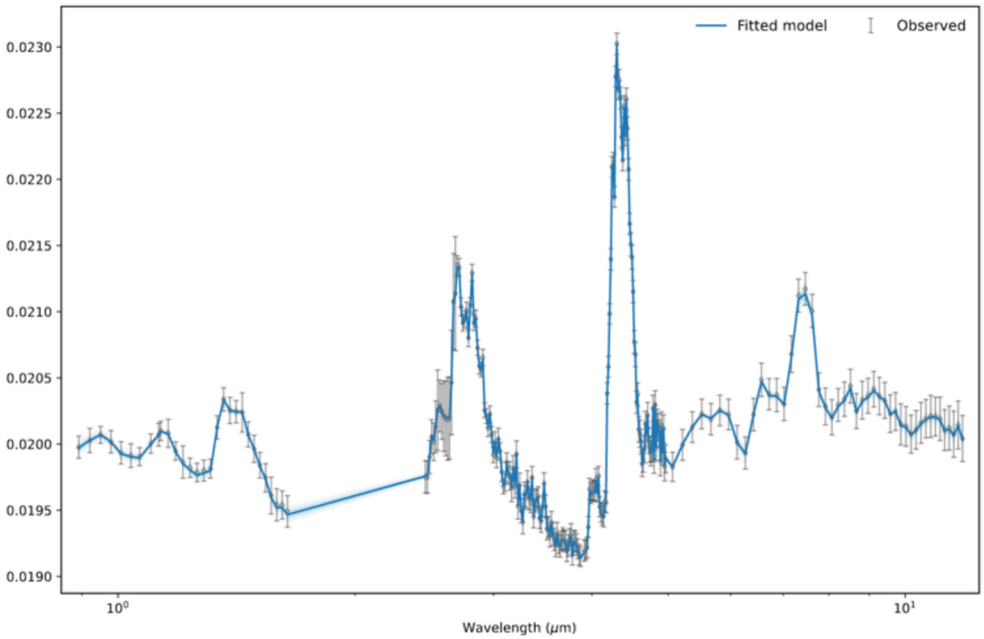
18 - Third iteration

Now the retrieval should look pretty neat with a great fit to the data, congrats!!



But... I forgot to tell you something. You see how the spectra are separated in two segments at 2 microns. I put them together in the *Tatooine.dat* file but is that right? They actually come from two different observatories, Hubble and JWST and have been taken years appart. Even if we somehow forget that the atmosphere of exoplanets varies with time (like the wheather on earth) nothing garanties us that the two spectra should be compatible. Luckily, with the *TauREx-instrumentsystematics* is perfect to deal with that kind of issues. This plugin adds parameters to the [Observation] header to enable retrievals on multi-telescope / multi-instrument datasets. The spectrum from each instrument is loaded separately and are applied different instrumental corrections:

- *path_spectra*: Path to spectra with the same format as native *TauREx*
- *offsets*: list of offsets of the same size as *path_spectra.*
- *slopes*: list of slopes of the same size as *path_spectra*.
- *broadening_profile*: path to files with the format: wavelength (um) spectral_resolution. The STScI .fits broadening profiles can also be used
- *error_scale*: list of multiplicative factors for the error bars.

For Tatooine, we are only going to use the *offsets* but I will also briefly describe the effects of the other parameters bellow. For the *offsets* parameters, *TauREx* simply apply a constant offset to the forward model before to compare it with the data in the retrieval process. These can be retrieve if the offsets are not known (usually the case). For instance, we do not know the offset of the HST and JWST Tatooine dataset. Then, let us put *offsets = 0 , 0* in the [Observation] header. Taking the JWST spectrum as our reference, we only need to retrieve O*ffset_1* which is the first value of the *offsets* list (hence the one applied to the HST data).

```
[Observation]
observation = spectra_instr
path_spectra = /path/to/Tatooine_HST.dat , /path/to/Tatooine_JWST.dat
offsets = 0 , 0
```

We only need to add the following to the [Fitting] section:

```
Offset_1:fit = True
Offset_1:prior = "Uniform(-0.003, 0.003)"
Offset_2:fit = False
```

From the figure that shows the spectrum and the retrieval fit 0.003 is the approximate height of the CO2 feature so there should be plenty enough room for the HST spectrum.

The *slopes* parameter acts like *offsets* but instead of a wavelength independent constant a slope is applied to the spectrum:
*flux = flux + slope*wavelength (+ offset if one is provided)*

The broadening profiles are needed when using JWST or ground based facilities as their line spread functions are designed to spread over several pixels. This result in the broadening of absorption lines compared to their theoretical widths. *TauREx-Instrumentsystematics* uses the instrument's broadening profiles to broaden the forward model so that is can be compared to observations.

Lastly, the *error_scale* parameter is a multiplicative factor that is only applied to the error bars. It is useful in JWST's context as the data calibration pipeline can underestimate error bars. If the error bar on the spectrum does not accurately reflect the error sources that can plague your data these systematics will impacts the retrieved atmospheric parameters. As for the *solpes* and *offsets* these can also be retrieved.